

1 Introduction to Regression

What Is Regression?

Regression is a statistical technique for modeling the relationship between a **response** (dependent) variable and one or more **predictor** (independent) variables.

Instead of treating events in a system as isolated, regression lets us ask: *if this variable changes, how does that one respond, on average?*

- Response variable: what we want to explain or predict (denoted y)
- Predictor variable(s): what we use to explain it (denoted x)
- The regression equation summarizes the average relationship between them

Why Regression?

- **Business:** does advertising spend increase sales, and by how much?
- **Medicine:** how does drug dosage affect blood pressure reduction?
- **Engineering:** how does curing temperature affect concrete strength?
- **Economics:** how does income relate to household spending?
- **Environment:** how does fertilizer amount relate to crop yield?

2 Simple Linear Regression (SLR)

The SLR Model Equation

For an outcome variable Y and a single predictor variable X , the model is given as

$$Y = \beta_0 + \beta_1 X + \epsilon, \quad \epsilon \sim N(0, \sigma^2)$$

- β_0 : the intercept — expected or average value of Y when the value of X is zero
- β_1 : the slope — the expected or average change in Y for a one-unit increase X
- ϵ_i : random error — everything affecting Y that X does not capture
- β_0, β_1 are unknown **population parameters**; we estimate them from data as $\hat{\beta}_0, \hat{\beta}_1$

The SLR Model in Terms of Observed Values

For a single predictor X (advertising spend) and response Y (sales), and observed data points (x_i, y_i) , $i = 1, 2, \dots, n$, the SLR model is

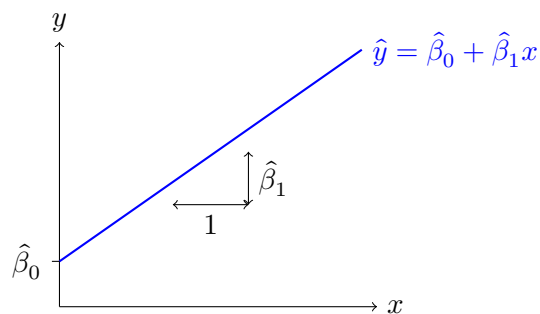
$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \quad \epsilon_i \stackrel{iid}{\sim} N(0, \sigma^2), \quad i = 1, \dots, n.$$

The fitted line is given by

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i.$$

What are the interpretations of $\hat{\beta}_0$ and $\hat{\beta}_1$ in this context?

Interpretation of the Parameters



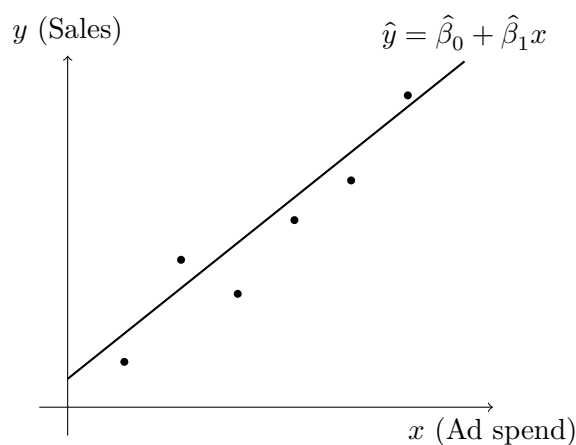
Assumptions of the SLR Model

- **Linearity:** the mean of y is a linear function of x , i.e. $E[\varepsilon_i] = 0$
- **Independence:** $\text{Cov}(\varepsilon_i, \varepsilon_j) = 0$ for $i \neq j$
- **Normality:** $\varepsilon_i \sim N(0, \sigma^2)$
- **Constant variance (homoscedasticity):** $\text{Var}(\varepsilon_i) = \sigma^2$ for all i
- x_i values are fixed, known constants, measured without error

3 Ordinary Least Squares (OLS) Estimation

The Regression Line

We want a line $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$ that fits the scatter of (spend, sales) points as closely as possible.

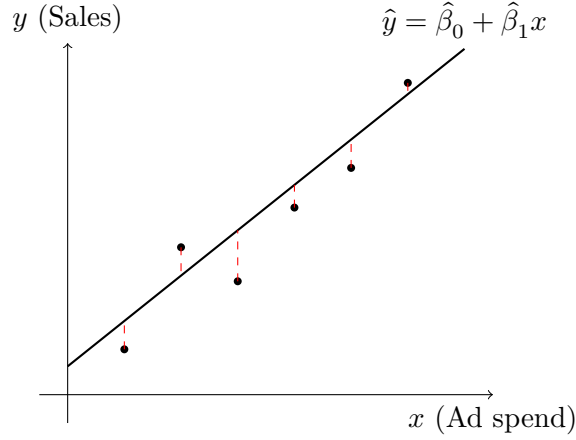


Residuals

For each data point, the **residual** is the vertical gap between the observed y_i and the fitted value $\hat{y}_i$:

$$\hat{\varepsilon}_i = y_i - \hat{y}_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i$$

Ordinary Least Squares (OLS) chooses $\hat{\beta}_0, \hat{\beta}_1$ to make these residuals collectively as small as possible.



Setting Up the Least Squares Problem

OLS minimizes the sum of squared residuals:

$$S(\beta_0, \beta_1) = \sum_{i=1}^n \varepsilon_i^2 = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$$

To minimize $S(\beta_0, \beta_1)$, we take partial derivatives with respect to β_0 and β_1 , set them equal to zero, and solve.

Since S is a quadratic (convex) function, this stationary point is the global minimum.

First-Order Conditions

Differentiate $S(\beta_0, \beta_1) = \sum_i (y_i - \beta_0 - \beta_1 x_i)^2$:

$$\frac{\partial S}{\partial \beta_0} = -2 \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)$$

$$\frac{\partial S}{\partial \beta_1} = -2 \sum_{i=1}^n x_i (y_i - \beta_0 - \beta_1 x_i)$$

Setting both equal to zero and evaluating at the minimizing values $\hat{\beta}_0, \hat{\beta}_1$ gives the **normal equations**.

The Normal Equations

$$\begin{aligned} \sum_{i=1}^n y_i &= n\hat{\beta}_0 + \hat{\beta}_1 \sum_{i=1}^n x_i \\ \sum_{i=1}^n x_i y_i &= \hat{\beta}_0 \sum_{i=1}^n x_i + \hat{\beta}_1 \sum_{i=1}^n x_i^2 \end{aligned}$$

This is a system of two linear equations in two unknowns, $\hat{\beta}_0$ and $\hat{\beta}_1$. We solve it directly using algebra.

Solving for $\hat{\beta}_1$

From the first normal equation, dividing by n :

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

Substitute into the second normal equation:

$$\sum_i x_i y_i = (\bar{y} - \hat{\beta}_1 \bar{x}) \sum_i x_i + \hat{\beta}_1 \sum_i x_i^2$$

Since $\sum_i x_i = n\bar{x}$, this becomes

$$\begin{aligned} \sum_i x_i y_i - n\bar{x}\bar{y} &= \hat{\beta}_1 \left(\sum_i x_i^2 - n\bar{x}^2 \right) \\ \hat{\beta}_1 &= \frac{\sum_i x_i y_i - n\bar{x}\bar{y}}{\left(\sum_i x_i^2 - n\bar{x}^2 \right)} \end{aligned}$$

Solving for $\hat{\beta}_1$ (cont.)

Define

$$S_{xy} = \sum_i (x_i - \bar{x})(y_i - \bar{y}) = \sum_i x_i y_i - n\bar{x}\bar{y}, \quad S_{xx} = \sum_i (x_i - \bar{x})^2 = \sum_i x_i^2 - n\bar{x}^2$$

$$\boxed{\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}}$$

Solving for $\hat{\beta}_0$

Having solved for $\hat{\beta}_1$, we plug back into the first normal equation:

$$\boxed{\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}}$$

Interpretation for our example: $\hat{\beta}_1$ is the estimated extra sales revenue per extra unit of advertising spend, and the fitted line passes through the point of averages $(\bar{x}, \bar{y})$.

OLS Estimators — Summary

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}, \quad \hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

Estimator of β_1 in terms of Covariance

Recall that

$$\text{CoV}(X, Y) = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{n - 1} = \frac{S_{xy}}{n - 1}$$

and

$$V(X) = \frac{\sum_i (x_i - \bar{x})^2}{n - 1} = \frac{S_{xx}}{n - 1},$$

hence,

$$\hat{\beta}_1 = \frac{\text{CoV}(X, Y)}{V(X)}.$$

Relationship between $\hat{\beta}_1$ and the Correlation Coefficient

Recall that the population correlation coefficient ρ is estimated by

$$r = \frac{\text{Cov}(X, Y)}{\sqrt{V(X) \cdot V(Y)}},$$

therefore

$$r = \frac{\sqrt{V(X)}}{\sqrt{V(Y)}} \hat{\beta}_1.$$

4 Numeric Example

Example: The Rocket Propellant Data

A rocket motor bonds an igniter propellant to a sustainer propellant inside a metal housing. The shear strength of that bond is a key quality characteristic, and it is suspected to depend on the age of the sustainer propellant batch.

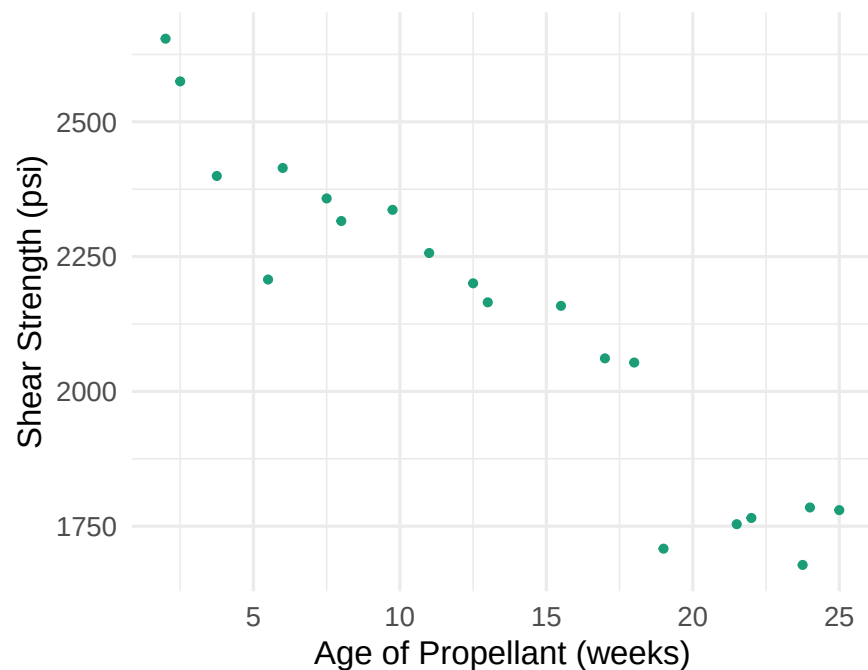
- $n = 20$ observations
- y_i : shear strength (psi)
- x_i : age of propellant batch (weeks)
- Tentative model: $y = \beta_0 + \beta_1 x + \varepsilon$

We will fit this model with OLS and check the algebraic properties from the previous frames against real numbers.

The Data ($n = 20$)

i	x_i (weeks)	y_i (psi)	i	x_i (weeks)	y_i (psi)
1	15.50	2158.70	11	13.00	2165.20
2	23.75	1678.15	12	3.75	2399.55
3	8.00	2316.00	13	25.00	1779.80
4	17.00	2061.30	14	9.75	2336.75
5	5.50	2207.50	15	22.00	1765.30
6	19.00	1708.30	16	18.00	2053.50
7	24.00	1784.70	17	6.00	2414.40
8	2.50	2575.00	18	12.50	2200.50
9	7.50	2357.90	19	2.00	2654.20
10	11.00	2256.70	20	21.50	1753.70

Scatter Diagram



Strength decreases as the propellant batch ages.

Computing S_{xx} , S_{xy} , and the Fitted Line

From the data: $n = 20$, $\bar{x} = 13.3625$, $\bar{y} = 2131.3575$.

$$S_{xx} = \sum_i x_i^2 - \frac{(\sum_i x_i)^2}{n} = 4677.69 - \frac{71,422.56}{20} = 1106.56$$

$$\begin{aligned} S_{xy} &= \sum_i x_i y_i - \frac{(\sum_i x_i)(\sum_i y_i)}{n} \\ &= 528,492.64 - \frac{(267.25)(42,627.15)}{20} = -41,112.65 \end{aligned}$$

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{-41,112.65}{1106.56} = -37.15$$

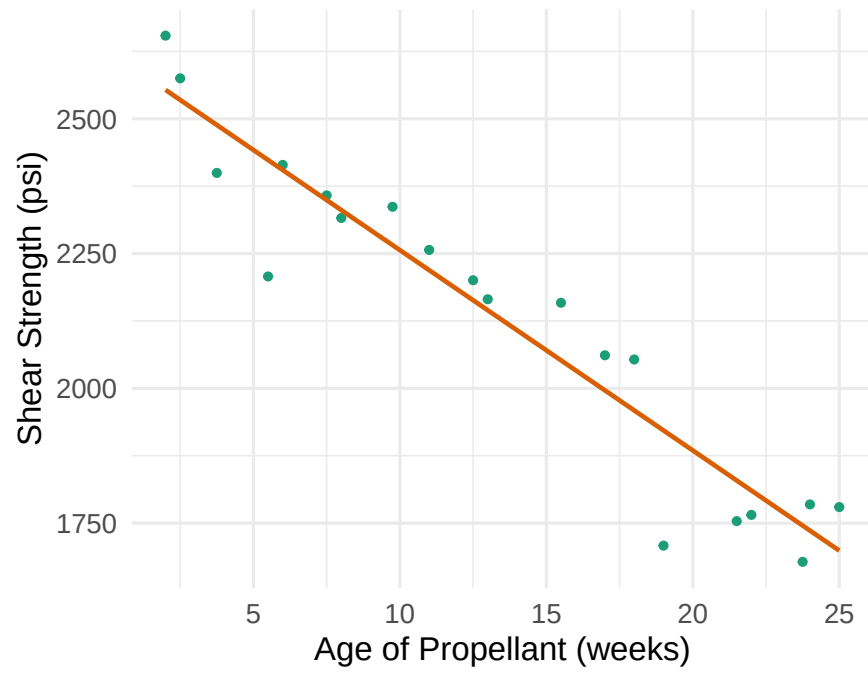
$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 2131.3575 - (-37.15)(13.3625) = 2627.82.$$

Fitted Equation

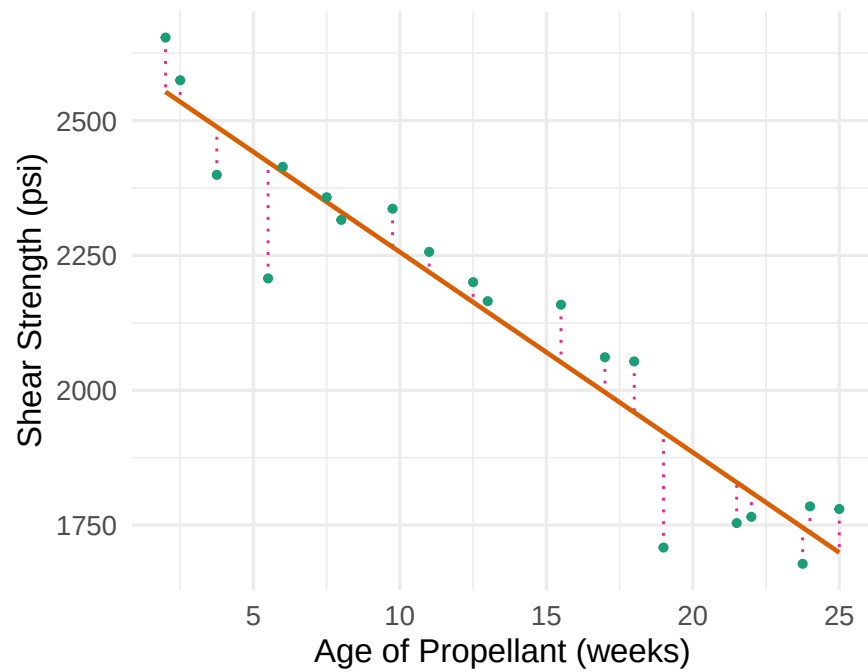
$$\hat{y} = 2627.82 - 37.15x$$

Each extra week of propellant age is associated with about 37 fewer psi of bond shear strength, on average.

Fitted Line



Fitted Line with Residuals



Fitted Values and Residuals

i	y_i	$\hat{y}_i$	$\hat{\varepsilon}_i$	i	y_i	$\hat{y}_i$	$\hat{\varepsilon}_i$
1	2158.70	2051.94	106.76	11	2165.20	2144.83	20.37
2	1678.15	1745.42	-67.27	12	2399.55	2488.50	-88.95
3	2316.00	2330.59	-14.59	13	1779.80	1698.98	80.82
4	2061.30	1996.21	65.09	14	2336.75	2265.58	71.17
5	2207.50	2423.48	-215.98	15	1765.30	1810.44	-45.14
6	1708.30	1921.90	-213.60	16	2053.50	1959.06	94.44
7	1784.70	1736.14	48.56	17	2414.40	2404.90	9.50
8	2575.00	2534.94	40.06	18	2200.50	2163.40	37.10
9	2357.90	2349.17	8.73	19	2654.20	2553.52	100.68
10	2256.70	2219.13	37.57	20	1753.70	1829.02	-75.32

Totals over all 20 observations: $\sum_i y_i = \sum_i \hat{y}_i = 42,627.15$, $\sum_i \hat{\varepsilon}_i = 0.00$.

5 Properties of the OLS Estimators

Rewriting $\hat{\beta}_1$ as a Linear Combination of the y_i

Since $\sum_i (x_i - \bar{x})\bar{y} = 0$, we can write

$$\hat{\beta}_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{S_{xx}} = \frac{\sum_i (x_i - \bar{x})y_i}{S_{xx}} = \sum_{i=1}^n c_i y_i, \quad c_i = \frac{x_i - \bar{x}}{S_{xx}}$$

The weights c_i satisfy two useful properties, both provable directly from the definition of S_{xx} :

$$\sum_i c_i = 0, \quad \sum_i c_i x_i = 1, \quad \sum_i c_i^2 = \frac{1}{S_{xx}}$$

These properties are needed to establish the unbiasedness of the estimators and find their variances.

Unbiasedness of $\hat{\beta}_1$

Treating the x_i as fixed and using $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$:

$$\begin{aligned} E[\hat{\beta}_1] &= E\left[\sum_i c_i y_i\right] = \sum_i c_i E[y_i] = \sum_i c_i (\beta_0 + \beta_1 x_i) \\ &= \beta_0 \underbrace{\sum_i c_i}_{=0} + \beta_1 \underbrace{\sum_i c_i x_i}_{=1} = \beta_1 \\ &\quad \boxed{E[\hat{\beta}_1] = \beta_1} \end{aligned}$$

On average, across repeated samples, the OLS slope estimate equals the true parameter.

Unbiasedness of $\hat{\beta}_0$

Using $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$ and $E[\bar{y}] = \beta_0 + \beta_1 \bar{x}$:

$$E[\hat{\beta}_0] = E[\bar{y}] - \bar{x} E[\hat{\beta}_1] = (\beta_0 + \beta_1 \bar{x}) - \bar{x} \beta_1$$

$$\boxed{E[\hat{\beta}_0] = \beta_0}$$

Both OLS estimators are unbiased.

Variance of the OLS Estimators

Because the ε_i are independent with common variance σ^2 , and $\hat{\beta}_1 = \sum_i c_i y_i$ with $\text{Var}(y_i) = \sigma^2$:

$$\text{Var}(\hat{\beta}_1) = \sum_i c_i^2 \text{Var}(y_i) = \sigma^2 \sum_i c_i^2 = \frac{\sigma^2}{S_{xx}}$$

A similar calculation for $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$ gives

$$\text{Var}(\hat{\beta}_0) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right)$$

More spread-out x values (larger S_{xx}) give more precise slope estimates — intuitively, advertising spend that varies a lot from month to month tells us more about its effect on sales than spend that barely changes.

The Gauss–Markov Theorem: OLS Is BLUE

Among all estimators of β_0 and β_1 that are:

- Linear in the y_i , and
- Unbiased,

the OLS estimators $\hat{\beta}_0, \hat{\beta}_1$ have the smallest possible variance.

This is why OLS is called **BLUE**: the **B**est **L**inear **U**nbiased **E**stimator. “Best” here means minimum variance, not that OLS makes no errors — it means no other linear, unbiased method can do better, under the SLR assumptions.

6 Other Useful Properties of the Least-Squares Fit

Other Useful Properties of the Least-Squares Fit

Beyond unbiasedness and BLUE, the OLS fit has several algebraic properties that follow directly from the normal equations — worth knowing since they show up in diagnostics and in software output.

- The residuals always sum to zero: $\sum_i \hat{\varepsilon}_i = 0$
- The observed and fitted values have the same sum: $\sum_i y_i = \sum_i \hat{y}_i$
- The fitted line always passes through the centroid $(\bar{x}, \bar{y})$
- Residuals weighted by x_i sum to zero: $\sum_i x_i \hat{\varepsilon}_i = 0$
- Residuals weighted by $\hat{y}_i$ sum to zero: $\sum_i \hat{y}_i \hat{\varepsilon}_i = 0$

Property 1 & 2: Residuals Sum to Zero

Recall the first normal equation from earlier:

$$\sum_i y_i = n\hat{\beta}_0 + \hat{\beta}_1 \sum_i x_i$$

Since $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$, summing over i gives

$$\sum_i \hat{y}_i = n\hat{\beta}_0 + \hat{\beta}_1 \sum_i x_i$$

The right-hand sides are identical, so

$$\boxed{\sum_i y_i = \sum_i \hat{y}_i} \implies \sum_i (y_i - \hat{y}_i) = \sum_i \hat{\varepsilon}_i = 0$$

This is not a coincidence — it is forced by the very equation OLS solves to find $\hat{\beta}_0$.

Property 3: The Fitted Line Passes Through $(\bar{x}, \bar{y})$

Divide the previous result $\sum_i y_i = \sum_i \hat{y}_i$ by n :

$$\bar{y} = \bar{\hat{y}} = \frac{1}{n} \sum_i (\hat{\beta}_0 + \hat{\beta}_1 x_i) = \hat{\beta}_0 + \hat{\beta}_1 \bar{x}$$

So the point $(\bar{x}, \bar{y})$ satisfies the fitted equation exactly — the **centroid** of the data always lies on the regression line.

This matches how we solved for $\hat{\beta}_0$ earlier: we built the line to pass through $(\bar{x}, \bar{y})$ by construction.

Property 4 & 5: Weighted Residuals Sum to Zero

Recall the second normal equation:

$$\sum_i x_i y_i = \hat{\beta}_0 \sum_i x_i + \hat{\beta}_1 \sum_i x_i^2$$

Since $x_i \hat{y}_i = \hat{\beta}_0 x_i + \hat{\beta}_1 x_i^2$, summing gives $\sum_i x_i \hat{y}_i$ equal to the same right-hand side, so

$$\sum_i x_i y_i = \sum_i x_i \hat{y}_i \implies \boxed{\sum_i x_i \hat{\varepsilon}_i = 0}$$

Property 4 & 5: Weighted Residuals Sum to Zero (cont.)

For the last property, write $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ and use both results just derived:

$$\sum_i \hat{y}_i \hat{\varepsilon}_i = \hat{\beta}_0 \underbrace{\sum_i \hat{\varepsilon}_i}_{=0} + \hat{\beta}_1 \underbrace{\sum_i x_i \hat{\varepsilon}_i}_{=0} = 0$$

$$\boxed{\sum_i \hat{y}_i \hat{\varepsilon}_i = 0}$$

7 Estimating the Error Variance σ^2

Residuals and the Error Sum of Squares

The parameter σ^2 measures how much sales vary around the regression line for a given level of advertising spend — it is unknown and must be estimated from the residuals.

Define the **error sum of squares**:

$$SS_E = \sum_{i=1}^n \hat{\varepsilon}_i^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

It can be shown that $E[SS_E] = (n-2)\sigma^2$. The divisor $n-2$ reflects that two degrees of freedom were used up in estimating β_0 and β_1 before we could compute residuals.

An Unbiased Estimator of σ^2

$$\hat{\sigma}^2 = MS_E = \frac{SS_E}{n-2}$$

Since $E[SS_E] = (n-2)\sigma^2$, it follows immediately that

$$E[\hat{\sigma}^2] = \sigma^2$$

so MS_E is an unbiased estimator of the error variance. This quantity underlies standard errors, confidence intervals, and hypothesis tests for β_0 and β_1 — topics for the next lecture.

8 Finding the Expected Value of SS_E

Why $E[SS_E] = (n-2)\sigma^2$: Rewriting the Residual

Recall $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x}$. Substituting into $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1x_i$:

$$\hat{y}_i = (\bar{y} - \hat{\beta}_1\bar{x}) + \hat{\beta}_1x_i = \bar{y} + \hat{\beta}_1(x_i - \bar{x})$$

So each residual splits into a piece explained by x and a leftover piece:

$$y_i - \hat{y}_i = (y_i - \bar{y}) - \hat{\beta}_1(x_i - \bar{x})$$

Why $E[SS_E] = (n-2)\sigma^2$ (cont.)

Squaring and summing over i :

$$\begin{aligned} SS_E &= \sum_i [(y_i - \bar{y}) - \hat{\beta}_1(x_i - \bar{x})]^2 \\ &= \underbrace{\sum_i (y_i - \bar{y})^2}_{S_{yy}} - 2\hat{\beta}_1 \underbrace{\sum_i (x_i - \bar{x})(y_i - \bar{y})}_{S_{xy}} + \hat{\beta}_1^2 \underbrace{\sum_i (x_i - \bar{x})^2}_{S_{xx}} \\ &= SS_E = S_{yy} - 2\hat{\beta}_1 S_{xy} + \hat{\beta}_1^2 S_{xx} \end{aligned}$$

Simplifying SS_E

So far:

$$SS_E = S_{yy} - 2\hat{\beta}_1 S_{xy} + \hat{\beta}_1^2 S_{xx}$$

Since $\hat{\beta}_1 = S_{xy}/S_{xx}$, we have $S_{xy} = \hat{\beta}_1 S_{xx}$, so

$$\hat{\beta}_1^2 S_{xx} = \hat{\beta}_1(\hat{\beta}_1 S_{xx}) = \hat{\beta}_1 S_{xy}$$

The last two terms combine into one:

$$-2\hat{\beta}_1 S_{xy} + \hat{\beta}_1 S_{xy} = -\hat{\beta}_1 S_{xy}$$

$$SS_E = S_{yy} - \hat{\beta}_1 S_{xy}$$

Expectation of S_{yy}

Subtracting $\bar{y} = \beta_0 + \beta_1 \bar{x} + \bar{\varepsilon}$ from $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$:

$$y_i - \bar{y} = \beta_1(x_i - \bar{x}) + (\varepsilon_i - \bar{\varepsilon})$$

Squaring and summing:

$$S_{yy} = \beta_1^2 S_{xx} + 2\beta_1 \sum_i (x_i - \bar{x})(\varepsilon_i - \bar{\varepsilon}) + \sum_i (\varepsilon_i - \bar{\varepsilon})^2$$

The x_i are fixed constants and $E[\varepsilon_i - \bar{\varepsilon}] = 0$, so the middle term vanishes in expectation. For the last term, using $E[\varepsilon_i^2] = \sigma^2$ and $\text{Var}(\bar{\varepsilon}) = \sigma^2/n$:

$$\sum_i (\varepsilon_i - \bar{\varepsilon})^2 = \sum_i \varepsilon_i^2 - n\bar{\varepsilon}^2 \Rightarrow E\left[\sum_i (\varepsilon_i - \bar{\varepsilon})^2\right] = n\sigma^2 - n \cdot \frac{\sigma^2}{n} = (n-1)\sigma^2$$

$$\boxed{E[S_{yy}] = \beta_1^2 S_{xx} + (n-1)\sigma^2}$$

Expectation of $\hat{\beta}_1 S_{xy}$

Using $S_{xy} = \hat{\beta}_1 S_{xx}$ again:

$$\hat{\beta}_1 S_{xy} = \hat{\beta}_1^2 S_{xx}$$

Since $E[\hat{\beta}_1] = \beta_1$ and $\text{Var}(\hat{\beta}_1) = \sigma^2/S_{xx}$:

$$E[\hat{\beta}_1^2] = \text{Var}(\hat{\beta}_1) + (E[\hat{\beta}_1])^2 = \frac{\sigma^2}{S_{xx}} + \beta_1^2$$

$$E[\hat{\beta}_1 S_{xy}] = S_{xx} \left(\frac{\sigma^2}{S_{xx}} + \beta_1^2 \right) = \sigma^2 + \beta_1^2 S_{xx}$$

The Final Result

Combining with the previous page:

$$E[SS_E] = E[S_{yy}] - E[\hat{\beta}_1 S_{xy}] = [\beta_1^2 S_{xx} + (n-1)\sigma^2] - [\sigma^2 + \beta_1^2 S_{xx}]$$

$$\boxed{E[SS_E] = (n-2)\sigma^2}$$

The $\beta_1^2 S_{xx}$ terms cancel exactly — that cancellation is why SS_E isolates leftover error variance rather than variance explained by the true slope.